Escape Times from Neural Network Basins: Delay and Heterogeneity, Not a Weibull Shape Law

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Abstract

Theories of flat-minima bias rest on mean escape-time formulas, yet the full distribution of how long an optimizer stays in a basin is rarely measured. We measure it: we run SGD and Adam from exact minima and true saddles of small tanh networks on FASHION-MNIST and DIGITS, across 165 conditions and 17,474 exits, and decide exit by converged quenching. Our simulator recovers the exact Kramers rate on a double well (median ratio 1.07). We test four numerical claims, and none holds as stated. Exponential times are rejected in 86% of conditions, but Weibull is the best model in only 4 of 28, its shape k lies in 0.7–1.3 in only 32%, and k is unrelated to curvature ($\rho = -0.06$). Most of the deviation from $k = 1$ comes from a start-up delay (refitting after early exits gives $k = 1.03$) and from mixing across minima (pooling gives $k = 0.52$–0.74). Doubling the batch size multiplies escape time by 5.1, not $\sqrt{2}$, while η/B still sets the time scale on FASHION-MNIST to within about 10%. Adam escapes about $2\times$ faster than SGD at matched in-basin loss, with no demonstrated dependence on basin width, but makes no exits at its default learning rate. Saddle exit follows the top eigenvector 82–86% of the time at low noise and 54% at high noise, and the eigenvalue gap does not predict that share. Escape is a discrete, delayed hop whose rate depends on noise and on the specific minimum.

1 Introduction

How long does an optimizer stay in a loss basin? Arguments for flat-minima bias, implicit regularization and learning-rate schedules all depend on the answer [Xie et al., 2021, Mori et al., 2022, Smith and Le, 2018]. Those arguments lean on mean escape-time formulas derived in asymptotic limits.

The mean is not enough. The full distribution tells us whether escape is memoryless, which is what the usual stochastic differential equation (SDE) picture assumes. In the high-barrier limit, Kramers theory predicts Poisson exits and therefore exponential escape times [Kramers, 1940]. Real optimizers run at finite noise, start at a specific point, and move through landscapes with many neighbouring minima. None of these conditions guarantees exponential times, and each one bends the distribution in its own way.

Prior work measures mean rates [Xie et al., 2021, Mori et al., 2022, Ibayashi and Imaizumi, 2021]. We found no study of distribution shape with censoring and model comparison in real networks, although keyword searches of arXiv are not proof of absence. Theory also disagrees on direction: Zhou et al. [2020] predict that SGD escapes a given basin faster than Adam, while a common intuition holds the reverse. On one hand, the SDE view gives clean scaling laws. On the other hand, few measurements test those laws against exact minima and saddles.

We test four numerical claims about escape. (H1) Escape times are Weibull, not exponential, with shape k in 0.7–1.3 depending on geometry. (H2) Adam escapes narrow basins 2–5$\times$ faster than SGD, with similar times at wide basins. (H3) Doubling the batch size B at fixed learning rate η multiplies mean escape time by $\sqrt{2}$, with B/η setting temperature. (H4) Hessian eigenvalue gaps

at saddles predict the escape direction with 70–85% accuracy. We fix the pass criteria before any experiment. We then run hundreds of independent replicas of real SGD and Adam from the same exact minimum or saddle of a small tanh network, and we decide exit by quenching each replica to the minimum it belongs to (see algorithm 1).

None of the four claims holds as stated, and each fails in an informative way. Exponential times are rejected in 86% of conditions, but only 32% of fitted k values lie in 0.7–1.3 (median 1.47), and k shows no relation to sharpness. Discarding the earliest quarter of exits moves median k from 1.47 to 1.03. Doubling B multiplies escape time by 5.1 [3.4, 8.0], about $\tau \propto B^2$, not $\sqrt{B}$. Adam is about $2\times$ faster at matched in-basin loss and slower by at least 7–34$\times$ at its default learning rate. At low noise, saddle exit follows the top eigenvector in 82–86% of runs, yet the gap does not predict that share.

Our contributions are:

- We build an escape-time instrument for real networks from exact critical points, converged quenching and interval-censored fits, and we validate it against the exact Kramers rate.
- We measure the shape of escape-time distributions across 28 well-observed conditions, and we decompose non-exponentiality into a start-up delay and heterogeneity across minima.
- We show that batch size acts far more strongly than a $\sqrt{B}$ law, and that the sign of the Adam–SGD comparison depends on learning-rate calibration.
- We show that saddle exit is highly reproducible even when it leaves the top eigenvector, which a linearized model cannot explain.

Section 2 reviews prior work, section 3 describes our method, section 4 reports results, and section 5 discusses them.

2 Related Work

Kramers escape. Kramers [1940] gives the overdamped rate over a barrier ΔE at temperature T as $\kappa = \frac{\sqrt{a|b|}}{2\pi} \exp(-\Delta E/T)$, with a and b the curvatures at the well and the saddle. At high barrier the exit is a Poisson process, so first-passage times are exponential ($k = 1$). Any deviation must come from outside that limit, such as a relaxation transient, a low barrier or a mixture of rates. We use this exact result to validate our instrument, and then ask which of these sources act in real networks.

SGD as an SDE. Mandt et al. [2017], Smith and Le [2018] and Jastrzębski et al. [2017] derive a temperature that scales as η/B. Zhu et al. [2019] show that anisotropic noise aligned with sharp directions speeds escape from sharp minima. With noise covariance close to the Hessian, Xie et al. [2021] find mean escape time exponential in $(B/\eta)\,\Delta L/H$. With noise proportional to the loss, Mori et al. [2022] find a rate that is a power law in the ratio $L(\boldsymbol{\theta}_s)/L(\boldsymbol{\theta}^\star)$, while $\log \tau$ stays linear in B/η. Ibayashi and Imaizumi [2021] prove exponential escape efficiency without a stationary distribution, and Şimşekli et al. [2019] argue for heavy-tailed noise, whose Lévy models give a polynomial dependence on barrier height. No theory above predicts $\tau \propto \sqrt{B}$. Unlike this line of work, which studies mean rates, we measure the distribution and its censored tail.

Adam versus SGD. Zhou et al. [2020] show with a Lévy SDE that, for the same basin, SGD has a smaller escape time than Adam (Kingma and Ba [2015]), because adaptive scaling enlarges the basin's Radon measure and smooths noise tails. Xie et al. [2022] separate the two parts of Adam: adaptive learning rates escape saddles efficiently but do not select flat minima as SGD does, while momentum helps pass saddles and hardly affects flat-minima selection. We found no direct measurement of an Adam-to-SGD escape ratio split by basin width. Any such ratio depends on how the two learning rates are matched, so we make that calibration explicit and report how the result changes with it.

Saddles. Jin et al. [2017] and Daneshmand et al. [2018] bound the rate of saddle escape under perturbation or intrinsic gradient noise, and neither claims which negative eigendirection wins. Fukumizu and Amari [2000] and Şimşek et al. [2021] show that unit duplication and symmetry create true saddles in small networks. We add a measurement of the winning direction at nonsymmetric saddles found by mode-following Newton, and we compare it with a linearized model that uses the measured noise covariance.

Table 1: The three landscapes. All networks have one tanh hidden layer of width h. Inputs for A and S are 24 PCA dimensions.

	Role	Data	h	Weight decay	Parameters	$\lambda_{\max}$
A	Main	FASHION-MNIST [Xiao et al., 2017], 2,000 examples	8	10^{-4}	290	0.62–5.48
B	Replication	DIGITS [Alpaydin and Kaynak, 1998], 1,797 examples	8	10^{-3}	610	0.34–0.58
S	Saddles	FASHION-MNIST, as A	3	10^{-4}	115	–

Algorithm 1 Exit check for one replica

Input: replica $\boldsymbol{\theta}_t$, minimum $\boldsymbol{\theta}^\star$, loss L, sharpness $\lambda_{\max}$
$q \leftarrow \max(1, \lambda_{\max})$; $\tilde{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}_t$ ▷ *work on a copy*
Relax $\tilde{\boldsymbol{\theta}}$ for $1{,}500\,q$ heavy-ball steps (lr $= 1/\lambda_{\max}$, $\beta = 0.9$)
if $\|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}^\star\| \le 0.3$ **then return** in basin ▷ *cheap gate*
Continue relaxing up to $5{,}000\,q$ total steps
$d \leftarrow \|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}^\star\|$; $b \leftarrow$ barrier on the segment from $\tilde{\boldsymbol{\theta}}$ to $\boldsymbol{\theta}^\star$; $\Delta L \leftarrow L(\tilde{\boldsymbol{\theta}}) - L(\boldsymbol{\theta}^\star)$
if $d > 0.05$ **and** ($b > 10^{-5}$ **or** $|\Delta L| > 2 \times 10^{-5}$) **then return** exited
return in basin ▷ *slow soft mode, not a new minimum*

Sharpness caveats. Keskar et al. [2017] link sharpness to large-batch training, and Dinh et al. [2017] show that sharpness is not invariant to reparameterization. We report $\lambda_{\max}$ and trace in one fixed parameterization and use them descriptively. Yaida [2019] give fluctuation-dissipation checks for stationarity, which motivates our attention to the transient that follows a start at $\boldsymbol{\theta}^\star$.

3 Method

Problem formulation. Let $\boldsymbol{\theta}^\star$ be a strict local minimum of a regularized loss L, with Hessian eigenvalues up to $\lambda_{\max}$. We start R independent replicas of an optimizer at $\boldsymbol{\theta}^\star$, with separate minibatch draws. The escape time T of a replica is the first step at which gradient flow from $\boldsymbol{\theta}_t$ no longer returns to $\boldsymbol{\theta}^\star$. We write τ for the Kaplan–Meier median of T. We compare the Weibull survival function with its rivals,

$$S(t) = \exp\big(-(t/\lambda)^k\big), \tag{1}$$

where $k = 1$ is the exponential law of Kramers escape, $k > 1$ means a rising hazard, and $k < 1$ means a falling hazard.

Models and data. We choose one-hidden-layer tanh networks with cross entropy plus L2, because tanh gives exact Hessians, and because L2 and underparameterization give strict minima with nonzero per-sample gradients. Table 1 lists the three landscapes. We take 12 minima on landscape A and 6 on landscape B, chosen evenly in $\lambda_{\max}$ rank before any escape measurement and Newton-polished in float64 ($\|\nabla L\| < 10^{-7}$, positive definite Hessian).

Optimizers. We run SGD and Adam (β values 0.9 and 0.999), with ablations for RMSPROP, SGD+MOMENTUM ($\beta = 0.9$, $\eta = 0.01$, same effective step as SGD at $\eta = 0.1$), and warm-started Adam. Each replica draws its own fixed-size minibatches. We use $R = 256$ at the base condition and 128 elsewhere (64 for default Adam), with a cap of 150,000 steps. The simulation runs in float32 on shared GPUs (about 29 worker-hours); fits and Hessians use the CPU.

What counts as an exit? At geometrically spaced checks (10% spacing, minimum gap 100 steps), we copy each replica and relax the copy by full-batch heavy-ball descent (algorithm 1). A replica has exited if the relaxed point converges to a different minimum. Times are interval-censored between checks.

This rule survived two failures, both found by testing it. First, short quenches mislead: a 400-step quench reported 0–34% exited where a converged quench reported up to 97%. Second, slow soft modes give false exits: a Newton audit found that 495 of the first 6,337 exits (7.8%) polished back to $\boldsymbol{\theta}^\star$, and in one minimum 71% of exits were false. We added the barrier-or-loss rule and repeated the affected runs with the same seeds. A barrier test alone would be wrong, because one DIGITS minimum has a real neighbour with no segment barrier.

Table 2: Verdicts on each numerical claim. Ranges in brackets are 95% confidence intervals. Unless noted, values are for FASHION-MNIST. k statistics cover 28 conditions.

Claim	Verdict	Key measurement
Weibull, not exponential	Partly	$k = 1$ rejected in 86%; Weibull best by AIC in 4 of 28
k between 0.7 and 1.3	Refuted	32% in range; median 1.47, maximum 3.33
k depends on geometry	Refuted	$\rho(k, \log \lambda_{\max}) = -0.06$, $p = .77$
Adam 2–5× faster at narrow basins	Partly	2.75× [1.89, 5.18]; 2.01× with medians observed
Adam similar at wide basins	Refuted	1.89× [1.24, 3.22]; no significant narrow/wide difference
Doubling B multiplies τ by $\sqrt{2}$	Refuted	×5.1 [3.4, 8.0]; slope 2.02 [1.64, 2.41], against 0.5
η/B controls temperature	Supported on FASHION-MNIST	Equal-ratio $\tau\eta$ within about 10%; DIGITS inconclusive (cap)
Top eigenvector predicts exit, 70–85%	Noise-dependent	54% at high noise, 82–86% at lower noise ($r = 1$)
Eigenvalue gap predicts accuracy	Refuted	ρ from -0.07 to $+0.23$, all n.s.

Experiments. E0 is a controlled 8-dimensional double well in the Gaussian minibatch-noise limit, with 1,000 replicas per condition. It validates the instrument against Kramers, and it does not model a network. E1 measures shape at all minima with SGD, $B = 16$, $\eta \in \{0.1, 0.2\}$. E2 compares Adam with SGD at Adam learning rates $\{0.004, 0.008, 0.016\}$ on A and $\{0.001, 0.002\}$ on B. We interpolate the median in $\log \eta$ to the per-minimum matched rate, where matched means equal mean excess full loss over steps 500–2,000, well before most exits. Matched rates are 0.0055–0.0109 on A and 0.00135–0.00152 on B. We define wide and narrow as the bottom and top thirds by $\lambda_{\max}$ ($n = 4$ each on A). E3 uses 6 (A) and 3 (B) minima with $B \in \{4, 8, 16, 32, 64\}$ at $\eta = 0.1$, plus the equal-ratio triple $(\eta, B) \in \{(0.05, 8), (0.1, 16), (0.2, 32)\}$. E4 uses 30 unique saddles from mode-following Newton ($\|\nabla L\| < 10^{-9}$), mostly index 2, with $R = 128$ and $(\eta, B) \in \{(0.1, 16), (0.02, 16), (0.1, 64)\}$. The winning eigenvector has the largest absolute projection at first crossing of radius r in the negative subspace.

Baselines. For shape, the baseline is the exponential law, tested against Weibull, gamma, lognormal and shifted exponential. For H3, the rival to a power law in B is the Arrhenius form with $\log \tau$ linear in B. For saddles, the baselines are chance (0.46 on average) and a linearized Monte-Carlo model with the measured per-sample noise covariance σ_i, in which component i grows as

$$z_i \, \sigma_i \, (1 + \eta |\lambda_i|)^t, \qquad z_i \sim \mathcal{N}(0, 1). \tag{2}$$

Statistics. We fit all models by interval-censored maximum likelihood. On simulated data, $k_{\text{true}} = 0.7, 1.0, 1.3$ gives 0.71, 1.03, 1.29, with 5% false rejection of $k = 1$. We apply Benjamini–Hochberg control at 5% to likelihood-ratio tests. Confidence intervals for k use 100 bootstrap replicates of replicas, and cross-minimum intervals bootstrap *minima*, the unit of generalization. When the median is not reached, we hold τ at the 150,000-step lower bound and flag it; Weibull extrapolations reached 10^9. Flagged medians are excluded from slope and factor fits. H1 analyses use conditions with at least 50% exits (28 of 36 E1 conditions). All intervals are 95%. Table 7 in appendix A lists the criteria we fixed in advance.

4 Results

The data cover 165 conditions, 23,104 replicas, 17,474 exits and 30 saddles. Table 2 gives the verdict on each claim, and the rest of this section gives the evidence.

4.1 Validation against Kramers

Can the instrument recover a known answer? Yes. At barrier-to-temperature ratio $\Delta/T \geq 4$, measured τ over the exact Kramers time is 1.03–1.13 (median 1.07) across three well widths (figure 1). Weibull k falls from 1.18 at $\Delta/T = 1$ to 1.00 at $\Delta/T = 6$ when starting at the minimum. A near-stationary start still gives 1.10 at $\Delta/T = 1$, so part of the low-barrier excess is not start-up delay.

4.2 What escape looks like

Exits are discrete hops. In a typical condition, a replica lands on one of a few specific neighbours (median 3 distinct destinations by loss, and the top destination takes a median 88% of exits). The

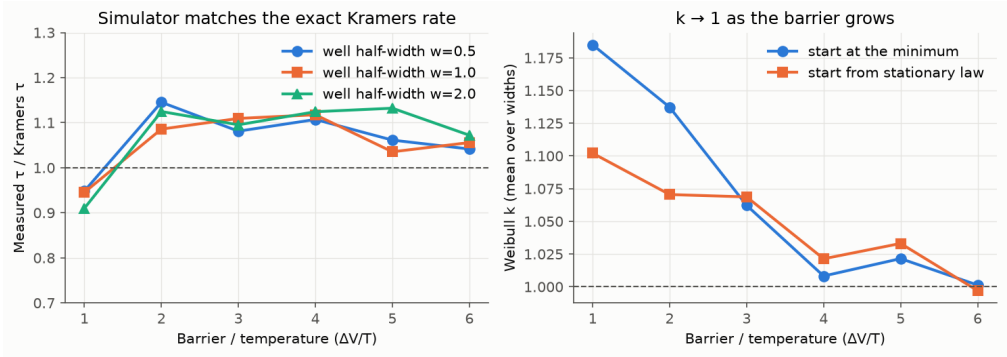


Figure 1: Validation on the controlled double well. *(Left)* Measured escape time over the exact Kramers time stays within 1.03–1.13 at $\Delta/T \geq 4$ for three well widths. *(Right)* Weibull k approaches 1 as the barrier grows, the Poisson limit Kramers theory predicts. A start at the minimum adds more excess k than a near-stationary start.

Table 3: Distribution shape across the 28 E1 conditions with at least 50% exits. Criteria failures in **bold**.

Statistic	**Value**	**Criterion**	**Met?**
Likelihood-ratio test rejects $k = 1$ (FDR 5%)	86%	majority	Yes
ΔAIC(Weibull $-$ exponential) < -10	75%	majority	Yes
k within 0.7–1.3	32%	$\geq$90%	**No**
$k < 1$ / $k > 1.3$	11% / 68%	both present	Yes
k range (median)	0.79–3.33 (1.47)	–	–
Spearman k vs $\log \lambda_{\max}$ / log trace / log det	-0.06 / -0.08 / -0.20	$p < .05$	**No** ($p = .77/.68/.31$)

median condition has 89% of exits landing at lower loss and 100% crossing a barrier. At the same (η, B), τ across minima spans 550 to more than 150,000 steps. A Newton audit of 192 accepted exits found 100% strict minima and 98.4% distinct from $\boldsymbol{\theta}^\star$ (residual false exits 1.6%, Wilson CI 0.5–4.5%). A same-seed GPU rerun of one condition reproduced every replica's exit check.

4.3 H1: distribution shape

Are escape times exponential? No (table 3). **Are they Weibull with k in 0.7–1.3?** Also no. The likelihood-ratio test rejects $k = 1$ in 86% of conditions, but only 32% of k values lie in the hypothesized range, and 68% exceed 1.3 (figure 2). Geometry does not explain the spread: k is uncorrelated with $\log \lambda_{\max}$, log trace and log det.

Weibull is also not singled out. The best AIC model is lognormal in 13 conditions, gamma in 8, Weibull in 4 and shifted exponential in 3. Across these 28 conditions, gamma beats Weibull by a median 5.1 AIC units and lognormal beats it by 6.4, while Weibull beats shifted exponential by 6.9. Adam shows the same pattern: on FASHION-MNIST, median k is 1.50 and 22% of the 32 conditions with at least 50% exits lie in range. Excluding the 8 conditions with fewer than 50% exits removes minima that barely escape, so we do not observe tail behaviour there.

Why $k \neq 1$. Two mechanisms account for most of the deviation. *Delay:* discarding the first 25% of exits and refitting residual time gives median $k = 1.03$ (IQR 0.90–1.20), and 50% of conditions fall within 0.85–1.15, against 18% before truncation. Discarding 50% gives 1.05. *Heterogeneity:* pooling minima gives $k = 0.74$ and 0.59 on FASHION-MNIST and 0.52 and 0.57 on DIGITS, against within-minimum means of 1.76, 1.51, 2.32 and 1.61. Pooling alone flips the sign of $k - 1$. Escape is still not fully memoryless after the delay. Hazards rise and then fall late (figure 3), and residual k still ranges from 0.74 to 1.79, so a constant hazard after a delay is only an approximation.

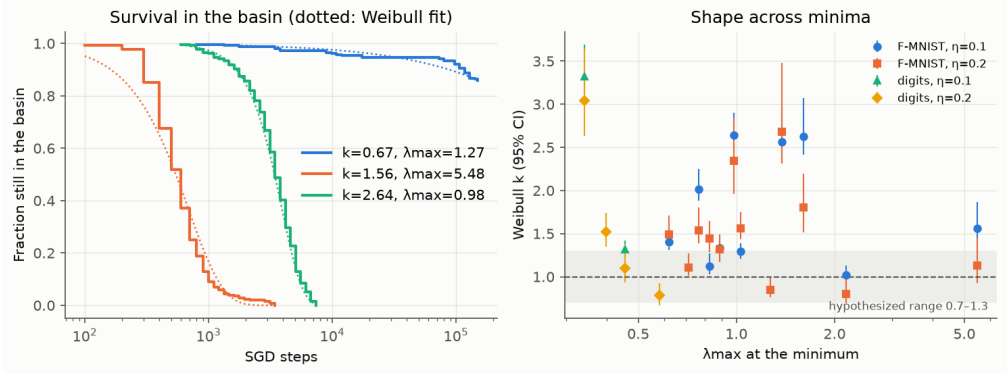


Figure 2: Distribution shape. *(Left)* Survival curves at three minima with Weibull fits (dotted). The $k = 0.67$ minimum has only 14% exits, so it is outside the 28 conditions in table 3. *(Right)* Weibull k against $\lambda_{\max}$. Most values lie above the hypothesized band, and no trend with $\lambda_{\max}$ appears.

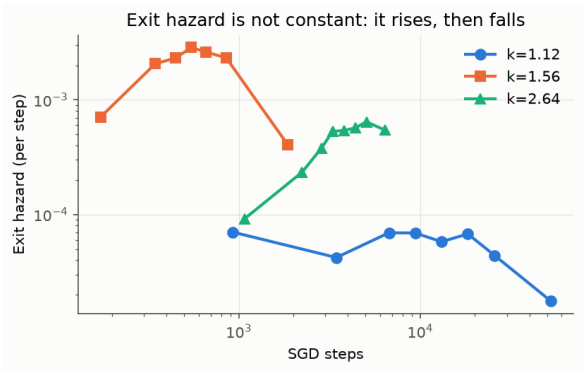


Figure 3: Exit hazard at three minima. The hazard rises during the early delay and falls late, including at minima with very different k. A Weibull with $k > 1$ cannot show the late fall.

4.4 H2: Adam against SGD

Is Adam faster at narrow basins only? No. At matched in-basin loss, Adam is about twice as fast across the board, with no demonstrated dependence on width (table 4, figure 4). The narrow point estimate of 2.75 is in the hypothesized range, but its interval [1.89, 5.18] is wider than [2, 5], so H2a fails its criterion. H2b fails because the wide ratio is 1.89 and its interval excludes 1. H2c is not significant: the narrow/wide ratio is 1.46 [0.74, 3.09], and the Spearman correlation of ratio with $\log \lambda_{\max}$ is 0.47 ($p = .12$). With $n = 12$ we cannot exclude a moderate effect. Keeping only minima with observed medians brings narrow and wide nearly level, at 2.01 and 1.91. On DIGITS the ratio is 1.56 [0.90, 3.22], with 5 of 6 minima involving a censored median, so this is weak evidence.

Ablations. RMSPROP reproduces Adam's speed-up without momentum, while SGD+MOMENTUM at the same effective η lands on the same median check as SGD at all 6 minima (resolution about 10%). So the adaptive scaling is the source, and warm-starting Adam shows that it is not a cold-start artefact. Calibration can reverse the direction: at its default learning rate of 0.001, Adam made no exits in 150,000 steps at 5 of 5 tested minima. That is at least 7–34× slower than SGD at the 4 minima where SGD's median is observed.

In the double well with theory-matched Adam, $\tau_{\text{SGD}}/\tau_{\text{Adam}}$ is 0.68–0.80 in narrow wells and 0.87–1.27 in wide wells. So Adam shows no consistent speed-up on a single barrier, even with curvature-aligned noise. The network speed-up is therefore not a generic property of adaptive scaling, and we did not test its cause.

Table 4: Adam against SGD on FASHION-MNIST. *Top:* $\tau_{\text{SGD}}/\tau_{\text{Adam}}$ at matched in-basin loss. The last column keeps only minima where all medians are observed; capped medians make the others bounds. *Bottom:* ablations on 6 minima, as τ/τ_{SGD} at $\eta = 0.1$. One of the 6 has a capped SGD median, so the second column drops it. Main estimates in **bold**.

Group	**Ratio, all minima**	**95% CI**	**Observed only**
Narrow (top third by $\lambda_{\max}$, $n = 4$)	**2.75**	1.89–5.18	2.01 ($n = 3$)
Wide (bottom third, $n = 4$)	**1.89**	1.24–3.22	1.91 ($n = 3$)
Narrow/wide ratio	1.46	0.74–3.09	–
All minima ($n = 12$)	2.42	1.83–3.27	1.97 [1.56, 2.53] ($n = 8$)
Ablation	**τ/τ_{SGD} (6 minima)**	**95% CI**	**Observed only**
RMSPROP, lr 0.008, no momentum	0.41	0.25–0.63	0.50 ($n = 5$)
Adam warm-started, lr 0.008	0.35	0.22–0.53	0.41 ($n = 5$)
SGD+MOMENTUM, same effective η	1.00	same median check	1.00 ($n = 5$)
Adam default lr 0.001 (5 minima)	No exits	–	$\geq$ 7–34$\times$ slower ($n = 4$)

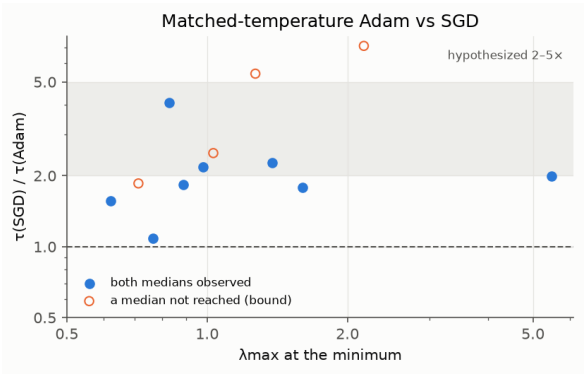


Figure 4: Adam speed-up against $\lambda_{\max}$ on FASHION-MNIST, at matched in-basin loss. Every minimum has ratio above 1, and the spread shows no clear trend with sharpness. Open markers involve a censored median and are bounds.

4.5 H3: batch size and temperature

Does doubling B multiply τ by $\sqrt{2}$? No. The factor is 5.1 [3.4, 8.0] on FASHION-MNIST, and the slope is 2.02 [1.64, 2.41] against the predicted 0.5, with per-minimum slopes of 1.5–2.8 (table 5, figure 5). The structural part of H3 does hold: within the observed range, a power law in B fits better than $\log \tau$ linear in B. This comparison needs care. It uses uncensored medians only, and Arrhenius predicts its fastest growth at large B, exactly where censoring removes data. The falling step factors could also reflect only fast minima surviving to large B. So our data favour a power law within the observed range, which is mostly $B = 4$–32, and they do not rule out Arrhenius behaviour in general. In E0, a single barrier, $\log \tau$ stays linear in $1/T$ even with loss-proportional noise, so that noise alone does not explain the network law.

Equal η/B. Is SDE time $\tau\eta$ invariant at fixed η/B? On FASHION-MNIST, approximately. Across the 5 minima with observed medians, $\tau\eta$ at $(0.05, 8)$ over $(0.1, 16)$ is 1.03 [0.97, 1.11], and at $(0.2, 32)$ over $(0.1, 16)$ it is 0.89 [0.86, 0.93], while raw steps differ by $2\times$ per pair. The second interval excludes 1, but the gap is about one check spacing, so SDE time is the right clock to within about 10%. Including the sixth minimum, capped at all three settings, gives 0.93 [0.72, 1.10] and 1.05 [0.87, 1.38]; these wider intervals partly reflect the ratios of the cap itself.

On DIGITS the test is close to uninformative, and we report it as inconclusive. All 3 medians at $(0.05, 8)$ are capped, so the reported ratio of 0.55 [0.50, 0.67] mostly reflects the cap. The ratio at $(0.2, 32)$ over $(0.1, 16)$ is 1.28 [1.02, 2.00], but only one minimum has both medians observed, and its ratio is 1.03.

Table 5: Batch-size scaling at $\eta = 0.1$. The hypothesis predicts slope 0.5 and factor 1.41. Lower AIC is better, and the preferred form is in **bold**. Fits use uncensored medians only.

Measure	FASHION-MNIST (6 minima)	DIGITS (3 minima)
Slope of $\log_2 \tau$ on $\log_2 B$	**2.02** [1.64, 2.41]	3.4 (one minimum)
τ factor per doubling	**5.1** [3.4, 8.0]	18 [10, 41]
Factor by step: $B = 4{\to}8$ / $8{\to}16$ / $16{\to}32$ / $32{\to}64$	5.9 / 3.8 / 4.2 / 2.4	23 / 5.1 / censored / censored
AIC: power law vs Arrhenius	$-$**16.7** vs $+16.1$	$-$**2.9** vs $+6.4$

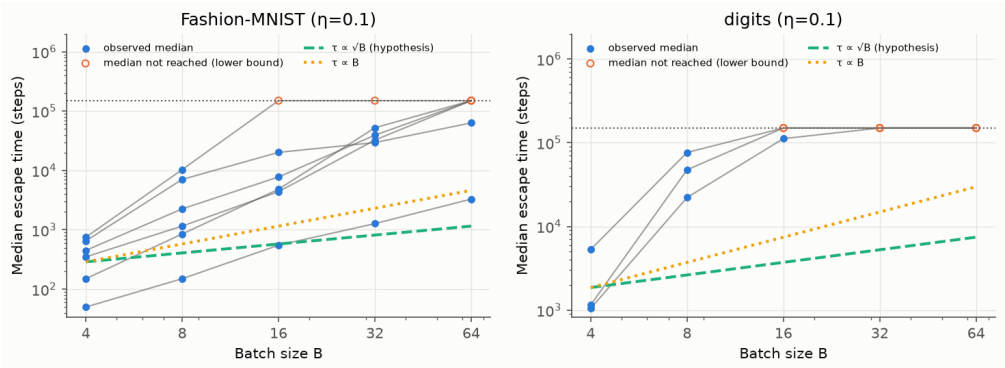


Figure 5: Median escape time against batch size at $\eta = 0.1$. Each gray line is one minimum. Measured growth far exceeds both $\tau \propto \sqrt{B}$ and $\tau \propto B$. Open markers at the dotted line are minima whose median exceeds the 150,000-step cap.

Doubling η at fixed B cuts τ by 9.4× [7.0, 13.4] on FASHION-MNIST. Two of the 12 minima enter this figure at the cap; the other 10 give a geometric mean of 8.5×. This agrees with 2× from time rescaling times the B factor above (4.2× at $B = 16 \to 32$).

4.6 H4: saddle exit direction

Does the top eigenvector win? It depends on noise (table 6, figure 6). At lower noise and $r = 1$, the share along $\boldsymbol{e}_1$ is 82–86%, which overlaps the 70–85% range. At high noise the share is 54%, and its interval includes chance. We did not fix the radius in advance, so we report all three. At $r = 1$ and low noise we dropped 3 of 30 saddles for too few crossings, which may flatter that cell.

Does the gap predict accuracy? No. Spearman ρ of the $\boldsymbol{e}_1$ share with relative gap is -0.07 (high noise), $+0.03$ (medium) and $+0.23$ (low), all non-significant ($p \geq .25$). For absolute gap, ρ runs from -0.24 to -0.07, also non-significant. Most saddles here have a large relative gap, so range restriction limits what a gap test can see.

At low noise the linear model of equation 2 is close (TV 0.13), and curvature matters more than noise projection (noise-only TV 0.30). At high noise, the model misses. One third of saddles have an $\boldsymbol{e}_1$ share below 20%, though the model predicts a median of 0.98. Per-saddle outcomes stay reproducible anyway, with modal share 88%. So exit there is near-deterministic along a non-$\boldsymbol{e}_1$ direction, which random linear growth cannot produce. Plausible causes are systematic noise-induced drift, or curvature that changes as thermal displacement moves the replica off the saddle. We tested neither.

5 Discussion

Does the SDE picture hold? In its scaling, largely yes: on FASHION-MNIST, $\tau\eta$ depends on η/B to within about 10%. In its usual exponential detail, not cleanly. A short-time delay, many exit channels and strong differences between minima dominate at these noise levels, while E0 shows that the instrument recovers Kramers when the theory's conditions hold. Sharpness, a common stand-in for escape difficulty, did not predict escape time here ($\rho = 0.08$ at $\eta = 0.1$ on FASHION-MNIST).

Table 6: Share of replicas exiting along the most-negative eigenvector $\boldsymbol{e}_1$, with 95% CIs over saddles (chance is about 0.46). TV is the total-variation distance from the measured exit distribution to the full linear model and to a noise-only model, at $r = 1$. Modal share is the share taken by each saddle's most common direction.

	Share along $\boldsymbol{e}_1$				**TV at $r = 1$**		
Noise (η, B)	$r = 0.1$	$r = 0.3$	$r = 1.0$	**Modal share,** $r = 1$	**Linear**	**Noise only**	$n, r = 1$
High $(0.1, 16)$	0.63 [.52, .74]	0.61 [.47, .75]	**0.54** [.39, .69]	0.88	0.46	0.42	30
Medium $(0.1, 64)$	0.65 [.53, .77]	0.68 [.54, .81]	**0.82** [.71, .91]	0.90	0.19	0.32	30
Low $(0.02, 16)$	0.66 [.54, .78]	0.71 [.58, .82]	**0.86** [.74, .96]	0.94	0.13	0.30	27

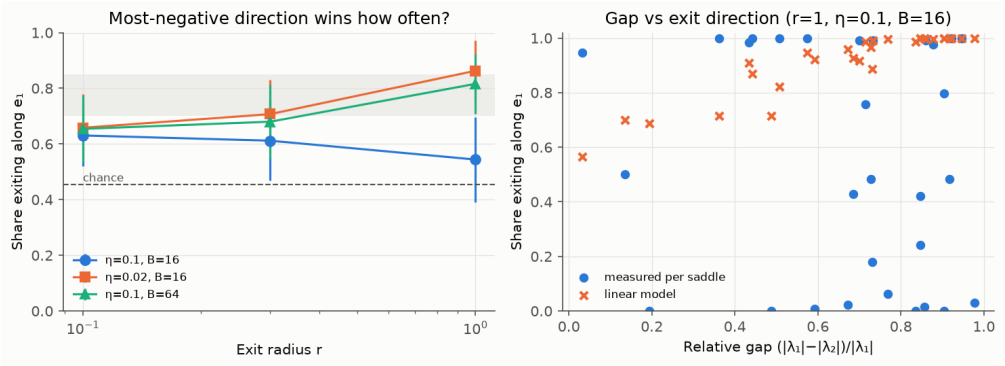


Figure 6: Saddle exit direction. *(Left)* Share along $\boldsymbol{e}_1$ against exit radius at three noise levels. The gray band is the hypothesized 70–85%. *(Right)* At high noise and $r = 1$, measured per-saddle shares (blue) split between near 0 and near 1 with no relation to the gap, while the linear model (orange) predicts near 1 for most saddles.

Alternative explanations. $k > 1$ *as a measurement artefact.* The converged quench and the Newton audit argue against it, and so does a robustness check: tripling quench length at 4 minima gives median ratios 1.00, 1.00, 1.10 and 0.64, and changes k by at most 0.16. The 0.64 is the sharpest minimum, where $\tau = 550$ spans only a few checks. *Delay as in-well relaxation.* This is consistent with the data, since excess loss relaxes over 1,000 or more steps, but we did not show it separately. E0 gives a smaller delay effect, with excess k of at most 0.2, so network delays are larger, possibly because of slow soft modes. *Adam's speed-up as calibration.* Matching excess loss is one reasonable choice, and the speed-up could partly reflect Adam's different stationary distribution. The reversal at the default learning rate shows how much calibration matters.

Relation to prior work. Our temperature scaling agrees with Smith and Le [2018] and Jastrzębski et al. [2017]. Steep dependence on B agrees qualitatively with Xie et al. [2021] and Mori et al. [2022], though the exp-linear form fits worse in our range. The claim of Zhou et al. [2020] that SGD escapes faster holds at Adam's default learning rate and reverses at matched loss, so the two views may differ mostly in calibration. The finding of Xie et al. [2022] that momentum matters little for escaping sharp minima fits our SGD+MOMENTUM ablation.

Practical implications. Use η/B as the control variable and expect a steep response: batch size has far more leverage on trapping than a $\sqrt{B}$ law suggests. Do not read Adam-versus-SGD escape claims without the learning-rate calibration. A single mean escape time also hides the delay and the spread between minima, so reports should give the distribution with censoring.

Limitations.

- *Scale.* We use tiny MLPs (115–610 parameters), small data and exact Hessians, and we show nothing for deep networks. Saddle results come from a different, smaller network than the minima results.
- *Minima sample.* We use 12 + 6 minima. Narrow versus wide comparisons have $n = 4$ per group and low power, and DIGITS has a small sharpness range.

- *Censoring.* The 150,000-step cap hides the slowest behaviour. It removes 8 of 36 E1 conditions from H1, turns many large-B medians into bounds, limits the Arrhenius test, and makes the DIGITS η/B test inconclusive.
- *Time resolution.* The check grid limits resolution below about 200 steps. Checks observe membership, so an exit and return between checks is missed.
- *Definition.* Results describe first exit from a strict gradient-flow basin, and a coarser "valley" definition would give different times. Residual false exits are about 1.6%.
- *Conditions.* An exact-replica start at $\boldsymbol{\theta}^\star$ builds in a transient. Two $\eta = 0.2$ runs at the sharpest minimum had $\eta\lambda_{\max} > 1$, where instability, not noise, may drive exit.
- *Saddle sample.* Saddles come from descent trajectories with a low-index filter, mostly index 2, and they have weak curvature ($|\lambda_1|$ from 2×10^{-4} to 0.03).

6 Conclusion

We measure escape-time distributions of SGD and Adam from exact minima and saddles of small tanh networks, and we test four numerical claims against criteria fixed in advance. Escape from these basins is a discrete hop with a non-constant hazard, shaped mainly by a delay within a minimum and by large differences between minima, not by a Weibull law with k in 0.7–1.3 tied to geometry. η/B is the right control variable on the main landscape, but τ scales about as B^2, not $\sqrt{B}$. Adam's advantage is real at matched in-basin loss (about $2\times$, from adaptive scaling and not momentum), is not shown to be specific to narrow basins, and disappears at the default learning rate. Saddle exit follows $\boldsymbol{e}_1$ at low noise, the gap does not predict how often, and at high noise exit is reproducible along other directions.

The main takeaway is that no single shape parameter summarizes escape: delay, minimum identity and noise level each change the answer. Future work should extend run lengths at large B to settle power law versus Arrhenius, start from conditioned quasi-stationary states to measure the delay directly, test the non-$\boldsymbol{e}_1$ saddle exits for noise-induced drift, and repeat these measurements on wider or deeper networks with Hessian-vector quenching. One open question remains: why does the hazard fall late within a single minimum, through competing channels or slow drift to harder regions?

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A Preregistered Criteria and Deviations

Table 7 lists the criteria we fixed before any experiment, with the outcome of each.

Table 7: Preregistered criteria and outcomes. Values are for FASHION-MNIST unless noted.

ID	Criterion	Outcome
H1a	LRT rejects $k = 1$ (FDR 5%) in a majority, $\lvert\Delta\text{AIC}\rvert > 10$	Met: 86% and 75%
H1b	k in [0.7, 1.3] in $\geq$90%, with both $k < 1$ and $k > 1$ present	Not met: 32%
H1c	Spearman $\rho(k, \log \text{sharpness})$ significant, $p < .05$	Not met: $p = .77$
H2a	Narrow $\tau_{\text{SGD}}/\tau_{\text{Adam}}$ CI inside [2, 5]	Not met: [1.89, 5.18]
H2b	Wide ratio CI inside [0.67, 1.5]	Not met: [1.24, 3.22]
H2c	Narrow/wide ratio-of-ratios > 1	Not significant: 1.46 [0.74, 3.09]
H3a	Slope CI contains 0.5, excludes 0 and 1	Not met: [1.64, 2.41]
H3b	Power law beats Arrhenius by AIC	Met in the observed range: -16.7 vs $+16.1$
H3c	Same η/B gives same $\tau\eta$ within CI	Partly met: 1.03 [0.97, 1.11] and 0.89 [0.86, 0.93] (5 minima)
H4a	Top-eigenvector accuracy CI overlaps [70, 85]% and exceeds chance	Met at lower noise ($r = 1$); not at high noise
H4b	Accuracy increases with gap	Not met: ρ from -0.07 to $+0.23$, n.s.

Deviations from the plan.

- We replaced the planned short quench with the converged quench, before production, for the reason in section 3.

- We added the barrier-or-loss rule after the Newton audit and reran the 10 affected runs with the same seeds.
- We moved the base condition from $\eta = 0.2$ to $\eta = 0.1$, because at $\eta = 0.2$ escape competes with in-well relaxation. We kept $\eta = 0.2$ as the hot condition.
- We made FASHION-MNIST at weight decay 10^{-4} the main landscape, because weight decay made DIGITS minima nearly uniform in sharpness.
- We used a smaller network for saddles, because mode-following Newton did not converge at 290 parameters. We rejected symmetric saddles, because per-sample gradients vanish along their escape directions.
- We saved but did not analyse the secondary loss-level and radius first-passage definitions, and we analysed quench-length and exit-rule robustness instead.
- 165 of 166 jobs finished. The missing job is default-learning-rate Adam at the sharpest minimum.
- We did not fix the H4 radius in advance, so we report all three radii and all three noise levels.